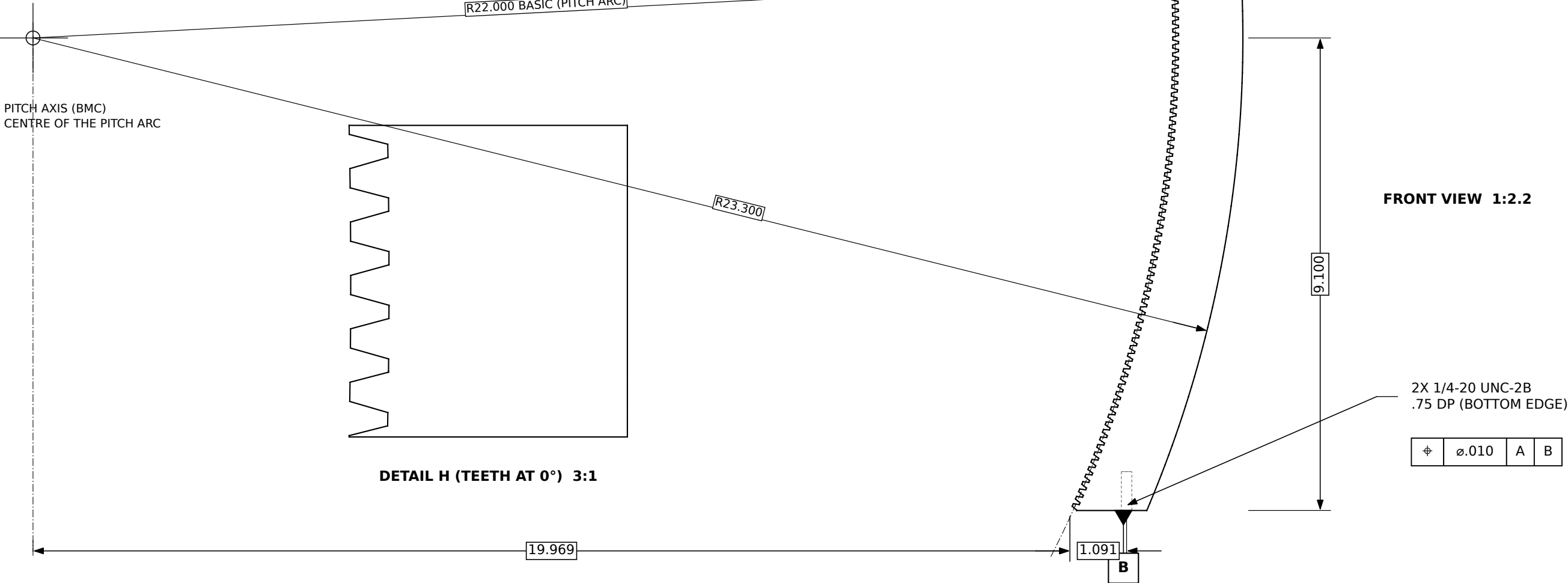


GEAR DATA

TYPE	INTERNAL ARC RACK (SEGMENT)
MODULE	1.25 mm
PRESSURE ANGLE	20°
PITCH RADIUS	22.000 in (558.8 mm) BASIC
TEETH ON SEGMENT	~87: FROM THE BASE TOP (-24.4°) TO +11°
ANGULAR PITCH	0.4026° BASIC
TOOTH THK AT PITCH	1.96 mm (-0.05/-0.10)
ADDENDUM / DEDENDUM	1.25 / 1.56 mm
MATING PINION	WT-6C-406, 20 T
QUALITY	AGMA Q7 / ISO 9



NOTES:

- TEETH: WIRE EDM (PREFERRED) OR CNC MILL WITH A MODULE 1.25 FORM CUTTER; PROFILE AND PROFILE FROM THE GEAR DATA TABLE. DEBURR TOOTH ENDS.
- THE PITCH ARC IS CENTRED ON THE PITCH AXIS (BMC), SAME ORIGIN AS THE CHEEKS (WT-6C-401): 9.100 ABOVE DATUM B.
- MATES THE 20-TOOTH PINION WT-6C-406. GRAVITY ALWAYS LOADS THE SAME TOOTH FLANK, SO BACKLASH DOES NOT REVERSE.
- FOOT: 2X 1/4-20 UNC-2B .75 DEEP INTO THE BOTTOM EDGE FOR THE BASE PLATE (WT-6C-403).

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXXX ±.0005

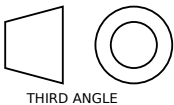
ANGLES ±0°30'

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 µin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



NEXT ASSY: WT-6C-400 PITCH SECTOR ASSY

PROJECT				DRAWN / DATE				
Wind tunnel - motorized 6-component sting balance				Claude (AI-generated) 2026-10-02				
TITLE				CHECKED		APPROVED		
RACK PLATE, PITCH (INTERNAL ARC RACK)								
MATERIAL				FINISH				
6061-T6 ALUMINUM, .500 PLATE				CLEAR ANODIZE TYPE II (TEETH INCLUDED)				
DWG NO.		REV	SIZE	SCALE		WEIGHT		SHEET
WT-6C-402		A	B	1:2.2 / 3:1		0.89 lb		26 OF 36